Train a Human Neural Network

Name _____

Work with a partner. Compare answers when you finish each problem.

The rule. total = ($A \times$ Attack weight) + ($S \times$ Special weight) + bias. More than 0: **FIGHTING**. 0 or less: **PSYCHIC**.

The fix rule. Right: no change. Said P, really F: **ADD** A, S, 1. Said F, really P: **SUBTRACT** A, S, 1.

Pass 1. Starting weights: Attack -1, Special +1, bias 0.

Card	A	S	Total	Says	Really	Right?	Attack w	Special w	Bias
T1 Machop	3	1			F				
T2 Abra	1	4			P				
T3 Machamp	4	2			F				
T4 Espeon	2	4			P				
T5 Hitmonlee	4	1			F				
T6 Kirlia	1	2			P				
T7 Conkeldurr	5	2			F				
T8 Alakazam	2	5			P				
T9 Makuhita	2	1			F				
T10 Mr. Mime	2	3			P				

Mistakes in pass 1: _____

Pass 2. Same cards, same order. Start with your weights from the end of pass 1.

Card	Total	Says	Right?
T1 Machop			
T2 Abra			
T3 Machamp			
T4 Espeon			
T5 Hitmonlee			

Card	Total	Says	Right?
T6 Kirlia			
T7 Conkeldurr			
T8 Alakazam			
T9 Makuhita			
T10 Mr. Mime			

Mistakes in pass 2: _____ (If you made mistakes, do a pass 3 on the back.)

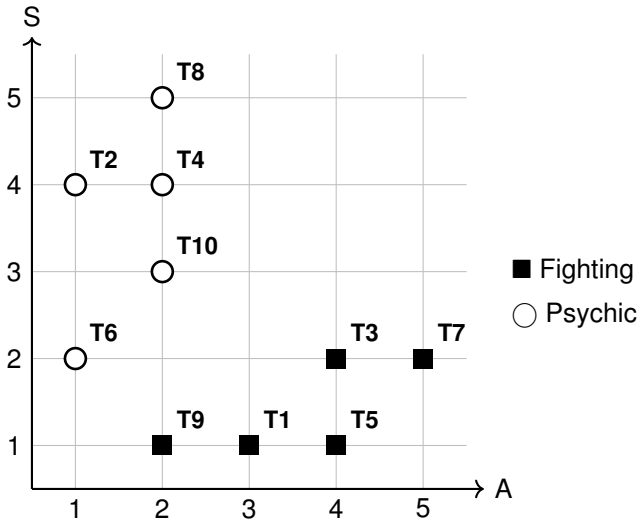
Test cards. Do **not** fix the weights on a test! Final weights: Attack _____ Special _____ bias _____

Card	A	S	Total	Says	Really (flip the card)
X1 Primeape	4	2			
X2 Mewtwo	4	5			
X3 Hypno	2	2			
X4 Gallade	4	2			
X5 Lucario	4	4			
X6 Medicham	2	2			

Talk about it. The neuron said Gallade is Fighting. Gallade is Psychic **and** Fighting. Was the neuron wrong?

The neuron **was** / **was not** wrong about Gallade, because _____

_____

_____

Go Further (Day 2). The neuron's line. The training cards are on the graph. Draw the line where the total is 0.